Upwind Bootcamp, Home Assignment

Gmail Add-on: Malicious Email Scorer

Objective

Design and implement a Gmail Add-on that analyzes an opened email and produces a maliciousness score accompanied by a clear, explainable verdict.

This task is intentionally open-ended. We are interested in how you approach an ambiguous problem, the trade-offs you make, and the quality of the solution you deliver, not in following a fixed specification.

What We Are Looking For

Your assignment will be evaluated on:

- **Product thinking**, which capabilities you chose to build, and why
- **Creativity**, the ideas you bring to the problem, and how you go beyond the obvious
- **Architecture and design**, how the components fit together, and the reasoning behind your decisions
- **Code quality**, readability, structure, and engineering hygiene
- **Security awareness**, how you handle untrusted input, secrets, and sensitive data
- **Communication**, how clearly you present your work in writing and in person

The solution does not need to be production-ready. We care more about clear thinking and clean execution than about feature completeness.

The Task

The core task is to build a **Gmail Add-on** that analyzes an opened email and presents a maliciousness score with a clear verdict to the user.

To get there, your solution should be structured as follows:

1. The Gmail Add-on is the product. It is what the user interacts with and what we will evaluate first.
2. Your solution should include a **backend service** that the add-on talks to. How you split work between the add-on and the backend is up to you.
3. The add-on should display a **clear and understandable result** to the user, including the score, the verdict, and the reasoning behind it.

Beyond these constraints, the scope, the capabilities you implement, the signals you analyze, and the way you design the experience are all up to you. Part of the assignment is deciding what is worth building.

Technical Guidelines

- The Gmail Add-on should be built using Google Workspace APIs.
- A backend service is expected. How and where you run it is up to you.
- The solution should be deployable to a real Gmail account and demonstrated live during the interview.
- Treat security as a first-class concern. Assume that emails, attachments, and external responses are untrusted input.

Deliverables

Your submission should be a **link to a public Git repository** containing:

1. **All source code** for the add-on and the backend.
2. **A `README.md`** at the root of the repository. The README is part of the evaluation, and we leave its contents to you. Treat it as a piece of professional communication. It should be the first thing a reviewer reads and should give them a complete picture of your work, including everything they need to understand it, run it, and assess the decisions you made.

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Demonstration

You will be asked to present your solution during the interview. Please bring your own laptop with the solution set up and ready to run, so the demonstration goes smoothly.

- Walk through the architecture and the reasoning behind your decisions.
- Demonstrate the add-on running against a real Gmail account.
- Discuss trade-offs, limitations, and what you would do differently with more time.

Good luck, and have fun with it.